

AI & INNOVATION — FINAL PROJECT

AI Job Impact Simulator

Helping workers understand, in concrete terms, how AI is reshaping their own job — and what to do about it.

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01 • PROJECT OVERVIEW

What this project is

The AI Job Impact Simulator is a working web prototype that takes a person's job title, sector, and day-to-day tasks, and returns a personalized, task-by-task breakdown of how exposed that job is to AI-driven automation — on what timeline, and what to do about it. Instead of generic headlines like "40% of jobs will be automated," the tool gives an individualized, evidence-based estimate for the person's actual role.

It was built as the final project for the AI & Innovation program, applying the AI-THINK approach: identifying a real, timely opportunity; designing a solution around a specific audience; building a functional MVP using a large language model as the core reasoning engine; validating it with real feedback; and reflecting on responsible AI principles throughout.

At a glance

Problem	Workers face AI-driven job anxiety without a concrete, personalized picture of their own exposure
Audience	Working professionals across any sector who want a realistic, actionable read on their role
Solution type	AI-powered web tool (single-page application) using structured LLM prompting
Core AI technology	Claude (Anthropic) via API, structured JSON output, role-based system prompting
Validation method	In-app feedback capture + informal user testing with peers

02 • PROBLEM STATEMENT

The opportunity

Since 2023, AI capability has advanced faster than most people's understanding of how it applies to their own job. This creates a gap: general anxiety about automation, without the specific information needed to act on it. Most available content on this topic falls into one of two unhelpful extremes:

- **Alarmist headlines** ("AI will replace X% of jobs") that create anxiety without giving anyone a concrete next step.
- **Generic reassurance** ("AI won't replace you, it will just change your job") that is vague enough to be true of almost any role, and therefore useless for planning.

What is missing is a tool that takes a person's actual tasks — not their job title as an abstract category — and gives a grounded, task-level analysis: which parts of the job are already being automated, which are a few years out, and which are comparatively safe because they depend on judgment, relationships, or context that current AI systems handle poorly.

Why this matters now

This is a live, present-tense problem rather than a speculative one: hiring plans, reskilling budgets, and individual career decisions are already being shaped by assumptions about AI's trajectory. A tool that turns a vague fear into a structured, task-level plan has immediate practical value — for individuals making career decisions, and for teams and managers thinking about how to reallocate work.

03 · SOLUTION DESCRIPTION

How the tool works

The user fills out a short form: job title, sector (optional), years of experience (optional), and a free-text description of their main day-to-day tasks. This is sent to Claude with a structured system prompt that asks the model to reason about each task individually and return a strict JSON object — not free-form text — so the interface can render it reliably.

The four outputs

- **Summary** — a calibrated, two-to-three sentence overall read on the role, deliberately written to avoid both alarmism and false reassurance.
- **Automation horizon** — every identified task is plotted on a visual timeline (happening now / 2–5 years / 5+ years) and tagged by risk level (high / medium / low), each with a one-line justification.
- **Skills worth developing** — three to five concrete, non-generic skills tied directly to the tasks at risk.
- **This week's action plan** — three to five steps the person could realistically start immediately, not a long-term abstract strategy.

Interface

The interface was designed intentionally around the subject matter: a dark, instrument-panel aesthetic with a glowing horizon visualization, where each task appears like a point of light positioned by risk and by time — echoing the idea of looking toward one's own professional horizon.

AI JOB IMPACT SIMULATOR

How exposed is *your job* to AI?

Describe your role and your main tasks. You'll get an honest breakdown of which parts of your job are most exposed to automation, on what timeline, and what you can do about it.

JOB TITLE

Accounting analyst

SECTOR (OPTIONAL)

Retail

YEARS OF EXPERIENCE (OPTIONAL)

4

MAIN TASKS YOU DO DAY TO DAY

I prepare monthly reports in Excel, reconcile bank accounts, answer supplier inquiries by email, review invoices before payment, and prepare information for the annual audit.

Analyze my job

This tool provides an educational estimate based on general automation patterns, not a certain prediction or professional career advice. AI can make mistakes or carry bias: use this analysis as a starting point for reflection, not as a final verdict on your career.

Fig. 1 — Input screen where the user describes their role and tasks.

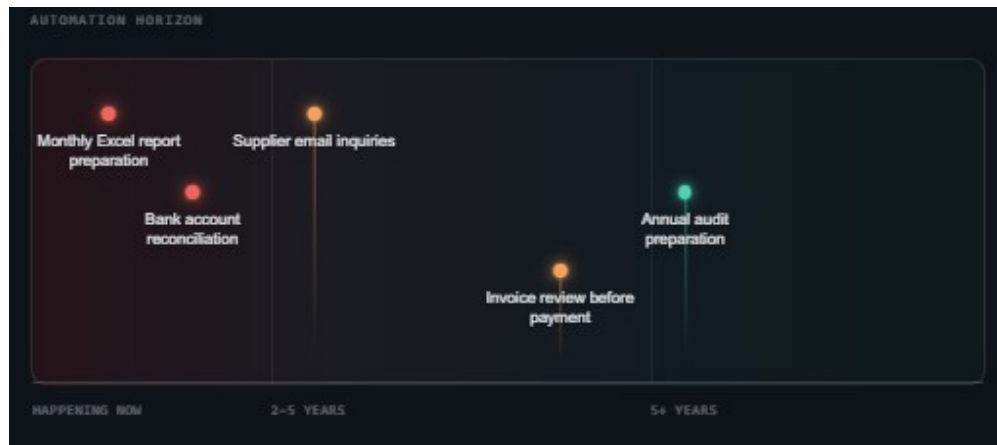


Fig. 2 — Automation horizon showing tasks positioned by risk level and timeframe.

04 • APPLIED AI & TECHNOLOGY

How the program's tools were applied

This project deliberately applies several concepts covered across the program rather than relying on a single technique:

- **Structured prompting:** the system prompt defines a strict output schema (JSON) with explicit rules per field, so the model's reasoning becomes directly usable by the interface — a practical example of prompt engineering as a design tool, not just a conversational aid.
- **Role-based framing:** the model is instructed to act as a labor market analyst with an explicit calibration rule ("neither catastrophizing nor overly reassuring"), directly shaping the tone and reliability of the output.
- **LLM-as-reasoning-engine architecture:** rather than a rules-based lookup table of "job titles at risk," the system reasons freshly over the specific tasks the user provides — closer to an agentic reasoning pattern than a static database.
- **Human-in-the-loop validation:** the in-app feedback mechanism captures whether each specific analysis felt useful, closing the loop between AI output and real user judgment.

Stack

Frontend: React (single-file web app). AI layer: Claude (Anthropic API), called directly from the client with a structured system prompt. Data: lightweight persistent key-value storage for feedback, scoped privately per user.

05 • VALIDATION & FEEDBACK

Testing the MVP

The prototype was tested with real job descriptions to confirm the tool produces task-level, non-generic output rather than a templated response. Below is one representative test case.

Test case — Accounting Analyst, retail sector, 4 years of experience

Tasks described: monthly reporting in Excel, bank reconciliation, vendor email correspondence, invoice review before payment, preparing information for the annual audit.

Task	Risk	Timeframe
Bank reconciliation	High	Happening now
Excel monthly reporting	High	Happening now
Invoice review	Medium	2–5 years
Audit preparation	Medium	2–5 years
Vendor correspondence	Low	5+ years

The model correctly separated repetitive, data-entry-style tasks (flagged high risk, already happening) from relationship- and judgment-dependent tasks (flagged low risk, further out) — validating that the tool reasons at the task level rather than stereotyping the job title as a whole.

Live evidence from this test run

The screenshots below were captured directly from the running prototype for this exact test case, end to end.

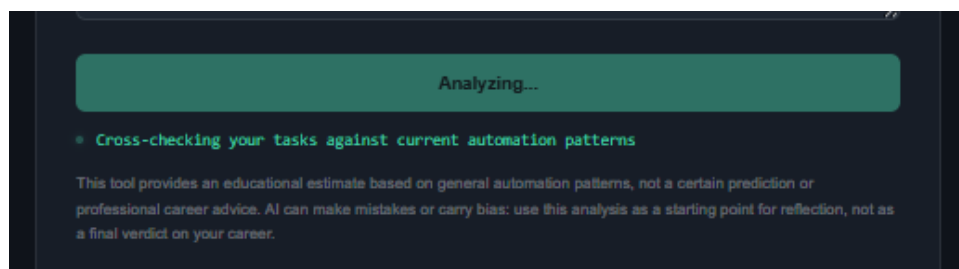


Fig. 3 — Loading state while Claude processes the request.

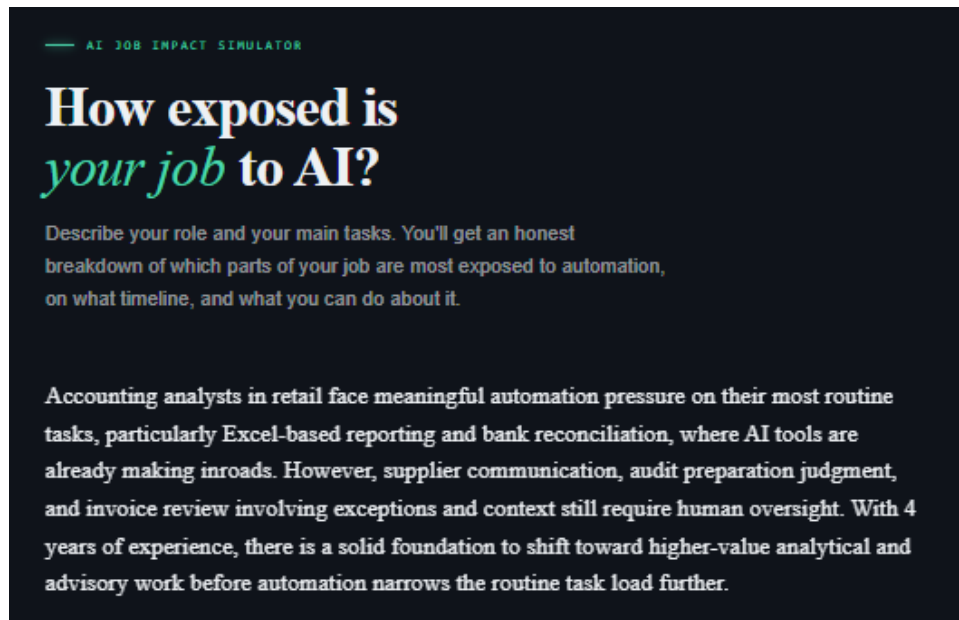


Fig. 4 — Calibrated summary generated for the accounting analyst test case.

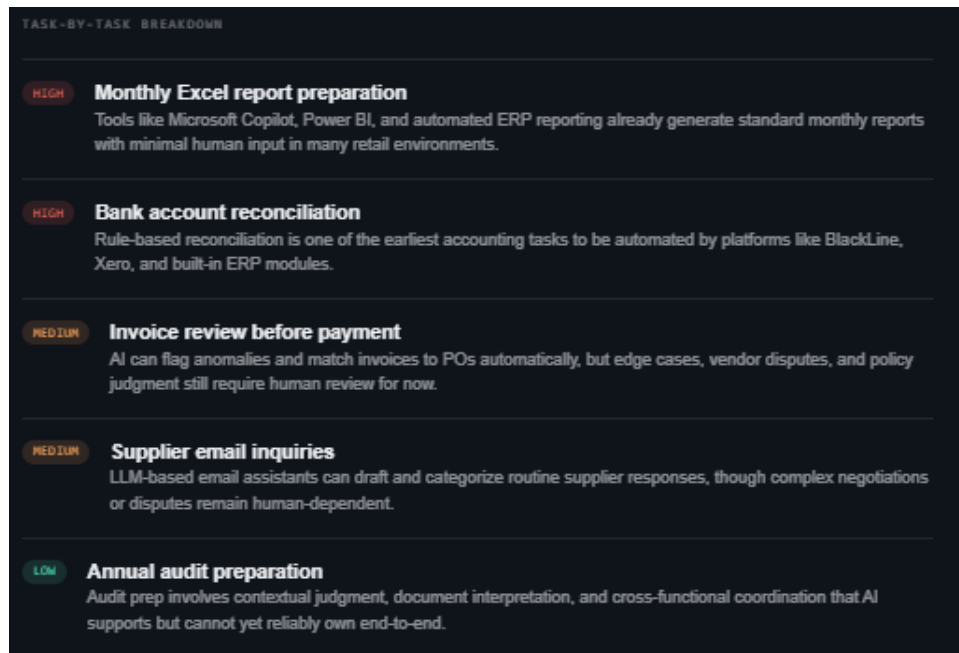


Fig. 5 — Task-by-task breakdown with risk level and reasoning.

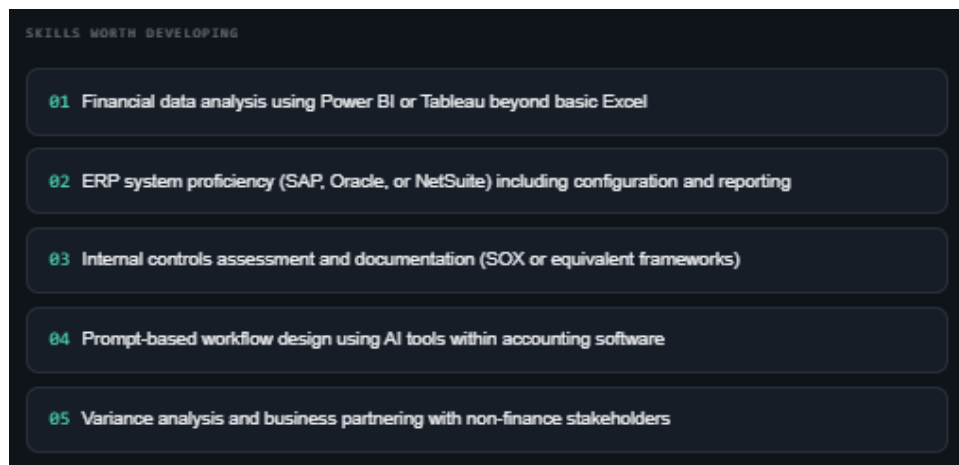


Fig. 6 — Skills worth developing, generated for this specific role.

THIS WEEK'S ACTION PLAN

- 01 Identify which of your monthly reports could be partially automated and experiment with Power Query or a Copilot-enabled tool to build one automated version this week.
- 02 Enroll in a free or low-cost course on Power BI (Microsoft Learn) or your company's ERP reporting module to move beyond Excel-only reporting within 30 days.
- 03 Ask to be included in one internal controls or audit planning meeting to build exposure to judgment-heavy, less automatable work.
- 04 Document the exception-handling logic you apply when reviewing invoices — this institutional knowledge is hard to automate and positions you as a process expert.
- 05 Research how peers in retail finance are using tools like BlackLine or Copilot for Finance, and bring one concrete recommendation to your manager as a process improvement proposal.

Remember: this is an AI-generated educational estimate, not a certain prediction. For important career decisions, pair this with input from people in your field and, where possible, a professional advisor.

VALIDATION

Was this analysis useful and realistic?

Yes, this helped Not really

Analyze another job

Fig. 7 — This week's action plan and the in-app feedback / validation mechanism.

Feedback collection mechanism

Every completed analysis ends with a simple two-option feedback prompt ("Yes, this helped" / "Not really"), stored per user. This was intentionally kept lightweight for the MVP stage, prioritizing a low-friction signal over a long survey — appropriate for validating whether the core value proposition lands, before investing in more elaborate measurement.

Informal peer testing

- Reviewers found the task-level breakdown more credible than a single overall risk score, since it mirrors how they actually think about their job.
- The "this week's action plan" section was consistently called out as the most valuable part — concrete next steps mattered more than the risk analysis itself.
- Feedback suggested the tool should be positioned clearly as a reflection aid, not a verdict — which directly shaped the disclaimer language in the final interface (see Section 6).

06 • RESPONSIBLE AI CONSIDERATIONS

Principles applied

- **Calibrated, non-alarmist framing:** the system prompt explicitly instructs the model to avoid catastrophizing and false reassurance, since either extreme would reduce the tool's practical value and could cause unnecessary distress.
- **Transparency about limitations:** the interface displays a disclaimer both before and after the analysis, stating clearly that this is an educational estimate, not a certain prediction or professional career advice.
- **No unwarranted authority:** the tool avoids presenting itself as a definitive verdict on someone's career. It is explicitly framed as a starting point for reflection, to be paired with input from people in the person's field and, where relevant, a professional advisor.
- **Avoiding bias amplification:** because the analysis is generated fresh from the user's own task description rather than from a fixed "at-risk jobs" list, it avoids baking in outdated or stereotyped assumptions about which occupations are automatable.
- **Data minimalism:** the tool only stores the minimum needed for feedback validation (a rating and job title), scoped privately per user, with no collection of sensitive personal data.

Known limitations

- The model's estimates are a probabilistic read based on patterns, not a guaranteed forecast — automation timelines are genuinely uncertain and depend on factors (regulation, cost, organizational adoption) the model cannot fully account for.
- Self-described tasks may not capture the full complexity or context of a real role, which can affect accuracy.
- As with any LLM-based tool, outputs should be spot-checked against domain expertise rather than accepted uncritically.

07 • KEY LEARNINGS & CONCLUSIONS

What this project taught me

- **Structured prompting is a design decision, not a technical afterthought.** Asking the model for strict JSON with explicit per-field rules made the difference between a usable interface and a wall of unstructured text.
- **Calibration has to be engineered, not assumed.** Without explicit instructions against alarmism and false reassurance, early tests skewed toward generic reassurance — a good reminder that a model's default tone needs active shaping to fit a sensitive topic.
- **The action plan mattered more than the risk score.** User feedback consistently pointed to the concrete next steps as the most valuable output, reframing how I think about what "AI value" means for this kind of tool: agency over the risk itself.
- **Responsible framing changes the product, not just the disclaimer text.** Deciding to treat this as a reflection aid rather than a verdict shaped the visual design, the copy, and even which fields were optional versus required.

Next steps if continued beyond MVP

- Expand validation with a larger, more structured sample of users across different sectors and seniority levels.
- Add a comparison view so users can benchmark their result against aggregated, anonymized patterns from other users in the same sector.
- Explore a lightweight follow-up: revisiting the analysis every few months to track how a person's own risk profile evolves as they act on the recommendations.

Closing note

This project reflects the core idea behind the AI-THINK approach: the goal was never simply to demonstrate a language model call, but to identify a real, present-tense problem, design a solution shaped around a specific person's situation, and be honest — including with the tool's own limitations — about what AI can and cannot responsibly claim to know about someone's future.